Bhagwan Arihant Institute Of Technology

Blockchain Technology

Internal Guide:- Abhay Hadiya
External Guide:- Gaurav Patel

Presented By:-
Patel Soham (201390107508)

Content

- Abstract
- Company Profile
- Work Plan
- Glimpse Of Training
- Training Summary
- Future Scope
- Conclusion
- Reference

Abstract

- The aim of this internship is to identify and describe the analysis carried out, the project completed, and the experience and knowledge gained as a Blockchain Developer at Advait Solutions. The analysis carried by a blockchain is essentially a distributed database of records or public ledger of all transactions or digital events that have been executed and shared among participating parties. That information can never be erased. My primary tasks are building decentralized applications (dApps) on the Ethereum blockchain, writing smart contracts in Solidity, and deploying them onto the blockchain using tools such as Remix and Truffle. This helped me to better understand how blockchain fits into the larger web development ecosystem and how it can be used to solve real-world problems. The experience at Advait Solutions has helped me become a more confident and capable developer with a strong foundation in blockchain.

Company Profile

- Company Name : - Advait Solution.
- Specialties and Expertise : - Database Management, Dynamic Website, CMS Development, PHP Based E-Commerce, API Integration, Cloud(Saas, Paas), CRM, Window Application, Web Designing, Application Development and Technology
- Advait Solutions is an IT Consulting & Software Development team delivering end to end IT solutions to Enterprises across the globe. With a team of 50+ passionate technology geeks spread across 3 countries, Advait Solutions is helping Enterprises overcome their business challenges, align their objectives with current IT trends and extract the maximum value for every dollar spent!

Work Plan

Monthly	week	Task
February	1st February to 10th February	• Introduction to Html, Form and Attribute • Introduction of CSS, JavaScript and basic task assignments • Introduction of Blockchain and Learning about Ethereum Network and crypto
	13th February to 17th February	• Introduction of Solidity and Remix • Learning about Blockchain Messenger, Boolean type, Mgs object, constructor • Introduction of Smart Contract
	20th February to 28th February	• Basic Transaction process of MetaMask • Introduction of Web3js • Basic Event of return variable in ERC20
March	1st March to 10th March	• Introduction of theory of Events, Modifier, inheritances in Blockchain • Learn about block Explorers and basic operation of source code verification in Etherscan • Introduction of Non-Fungible-Token (NFT) • Understanding of Truffle and Installation of truffle prerequisites
	13th March to 17th March	• Starting new truffle project, migration and testing of smart contract • Learning about debugging of smart contract • The Fundamental used of console of solidity ganache and debug is use of chain forking in smart contract
	20th March to 31th March	• Learn about Hardhat and installation of hardhat and the deploying smart contract to goerli and unit testing of hardhat • Understanding about foundry and forge and the knowledge about how the contract is adding a NFT, understand virtual machine pranking in solidity • Learning and create a To-do list min project and deploy the smart contract using MetaMask

Table 1 Work plan

Work Plan

Monthly	week	Task
April	3rd April to 12th April	• Learning about slides and labs, To Provide LAB addresses and Global Mgs object for smart contract
		• Learning of interacting with smart contract using Web3js and learn about RPC IPC WebSocket connection
		• Task 1 – Transaction of Ether between tow account
	13th April to 25th April	• Project 1
		• Project 2

Table 2 Work plan

Glimpse of Training

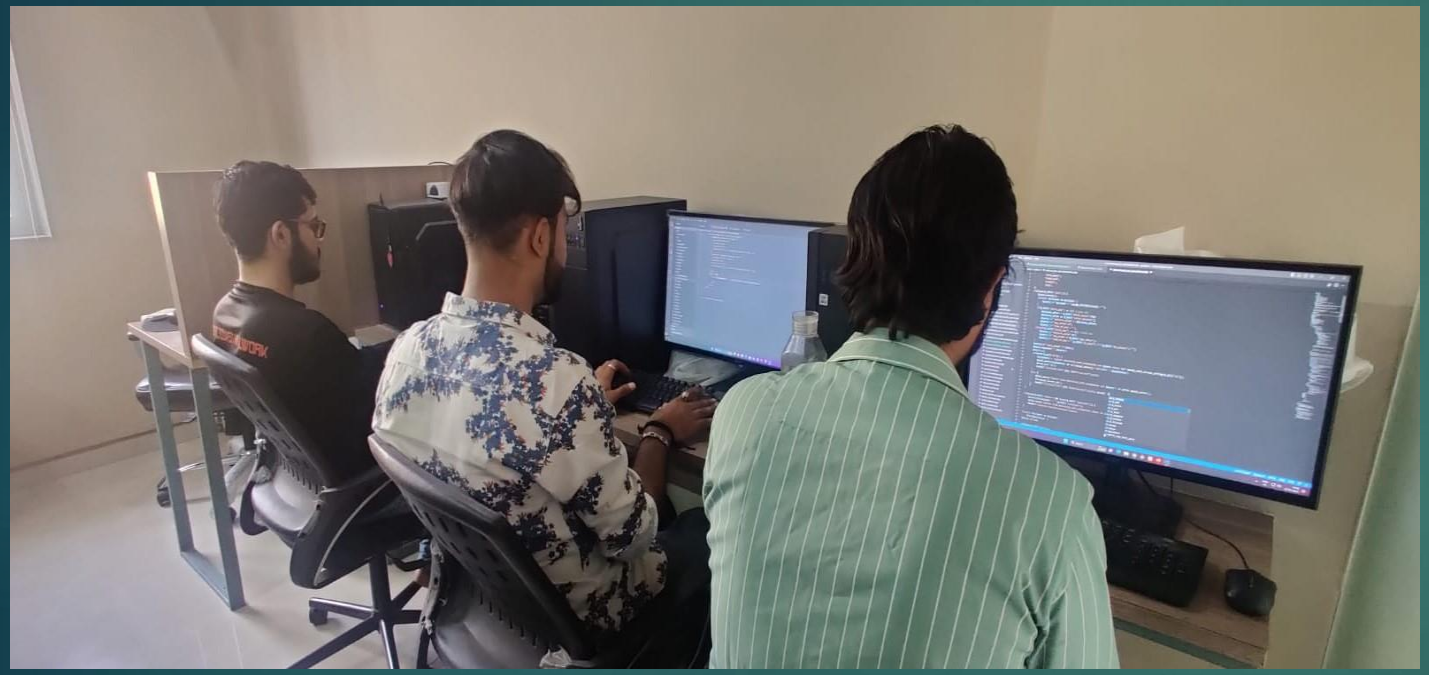

Fig. 1 Performing Task

Fig. 2 Office Area

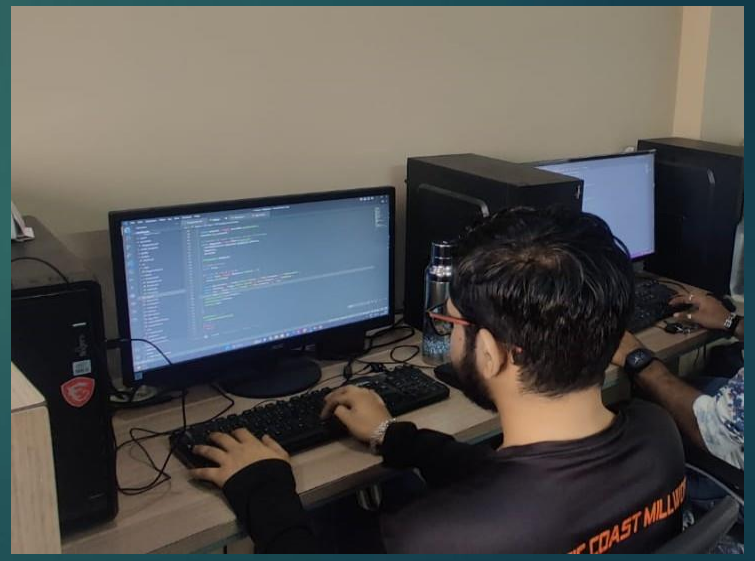

Fig. 3 Performing Task

Training Summary

- The journey began with an introduction to the company's operations and its unique work culture. The training program commenced with an in-depth study of blockchain technology, starting from the fundamentals and moving on to advanced topics. As part of the training, I delved into the intricacies of Solidity programming language and smart contract development. During the Solidity training module, I learned about the process of deploying a smart contract and how to seamlessly interact with it. I also honed my skills by designing and implementing several miniature projects using Solidity on the Remix platform. Additionally, I started to explore the world of decentralized applications (DApps) and web3 applications by leveraging my newfound knowledge of Solidity. Overall, the training journey was a thoroughly enriching and insightful experience that has equipped me with the necessary skills to succeed in the blockchain industry.

Remix interface for smart contract development

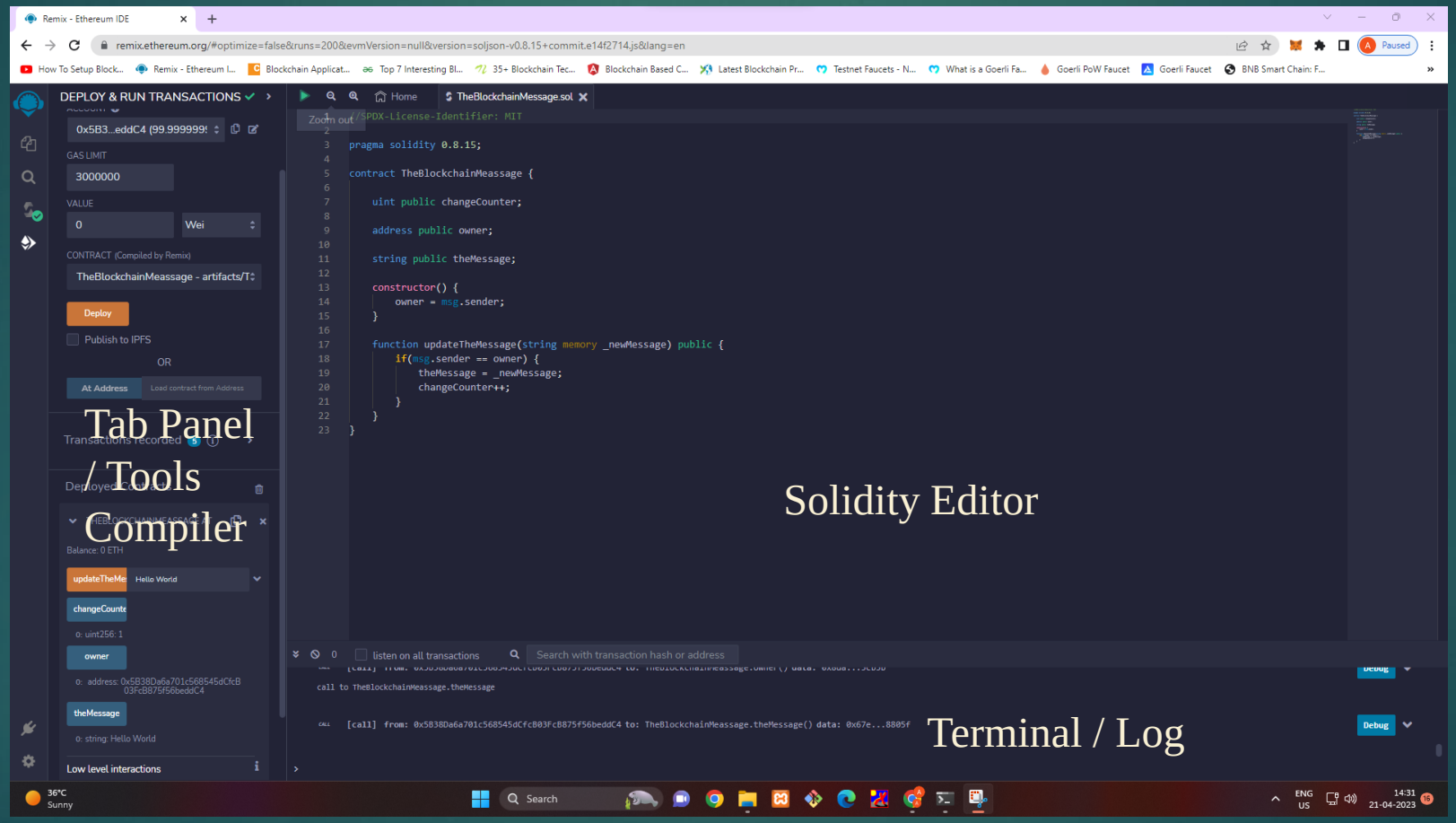

Fig. 4 Interact with smart contract Example

Deployment section

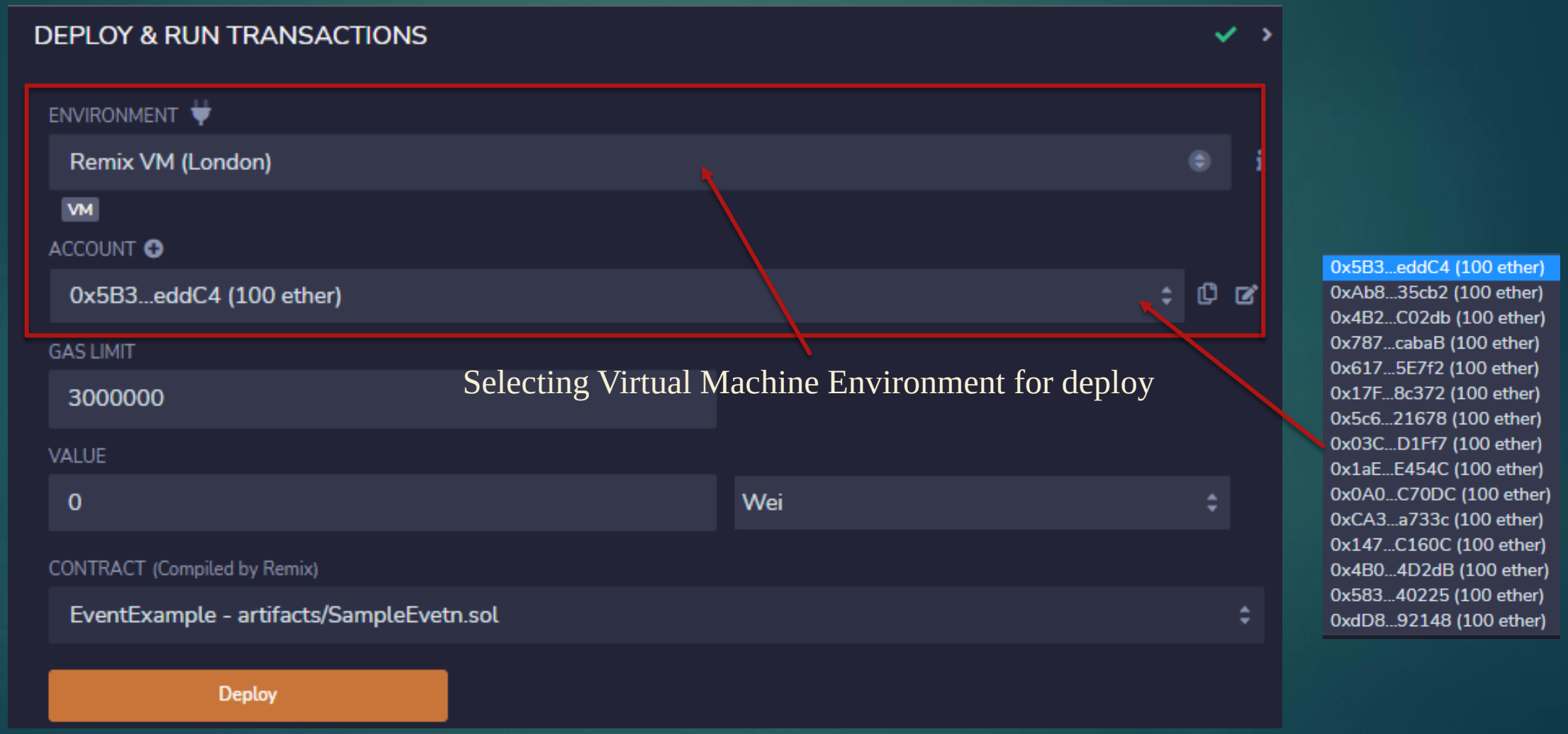

Fig. 5 Deploy and Transaction section

Deploy Contract For Interaction

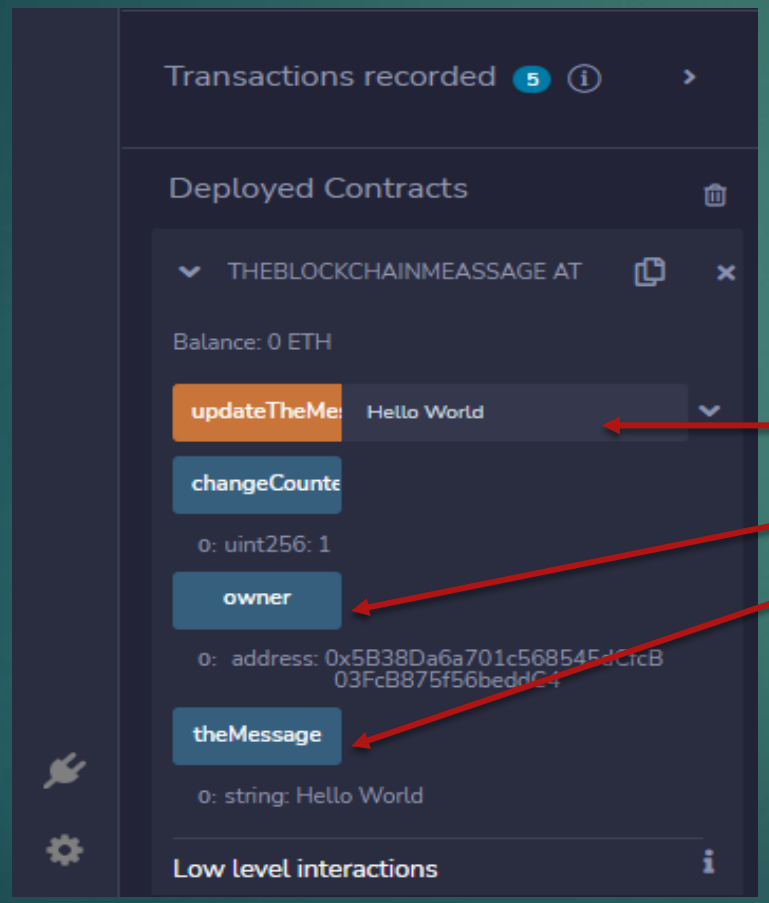

Fig. 6 Interact with Contract

Interact with the deployed contract:

1. Enter message and update it.
2. Show the owner who pass the message.
3. Display the message.

Smart Contract of To-Do list

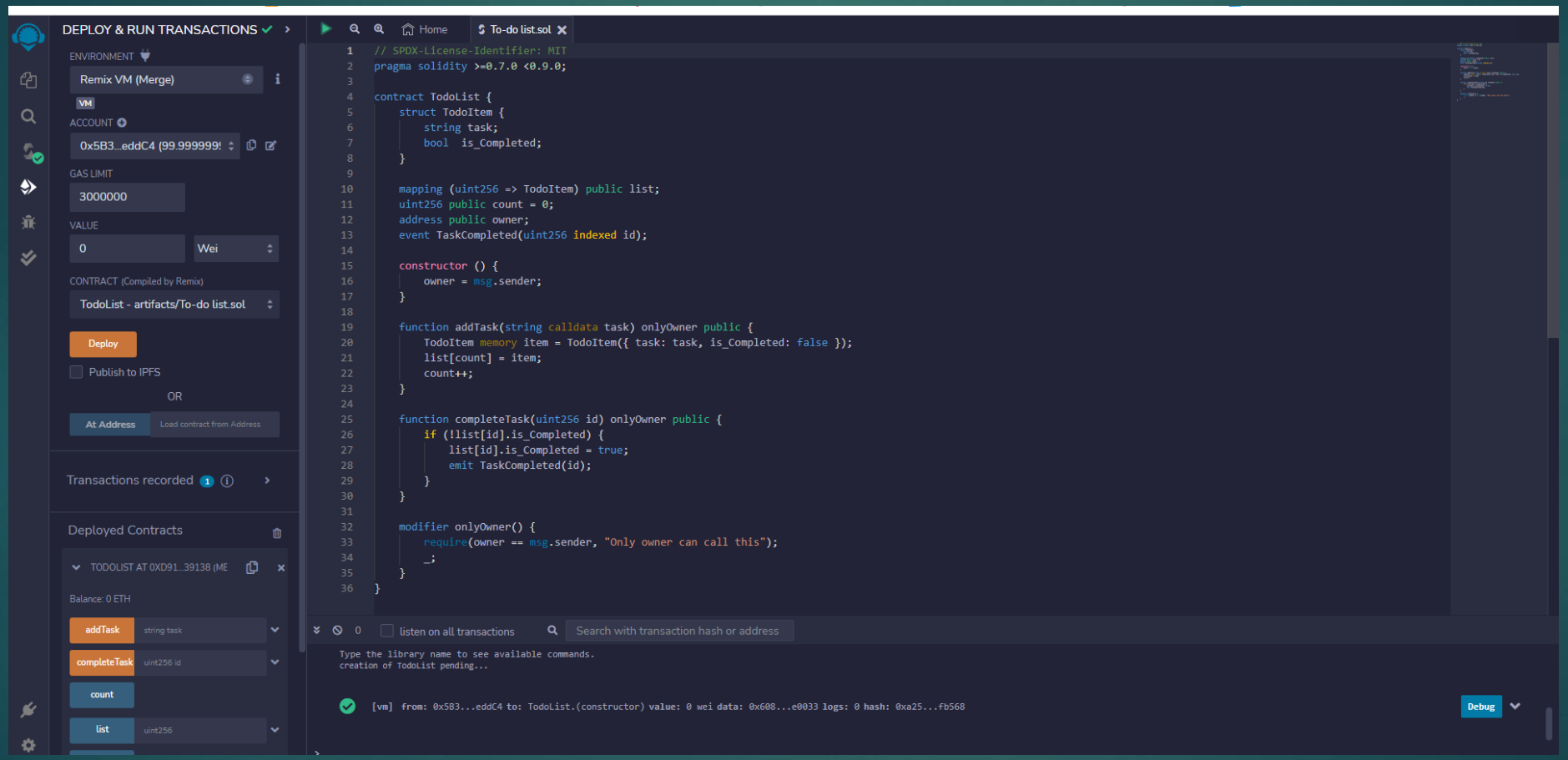

Fig. 7 To-Do list Mini project

Interact with Smart Contract of To-Do list

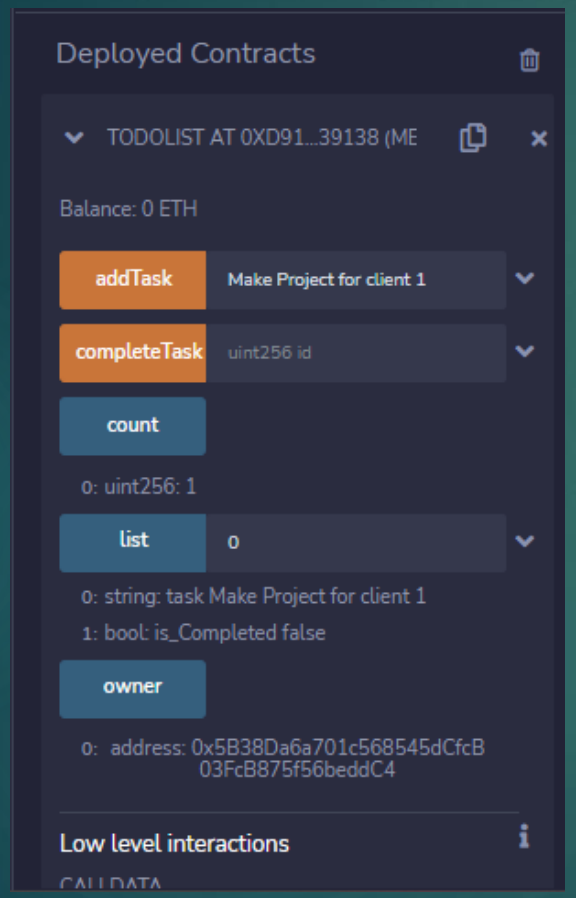

Fig. 8 Deploy and interact with To-do list smart contract

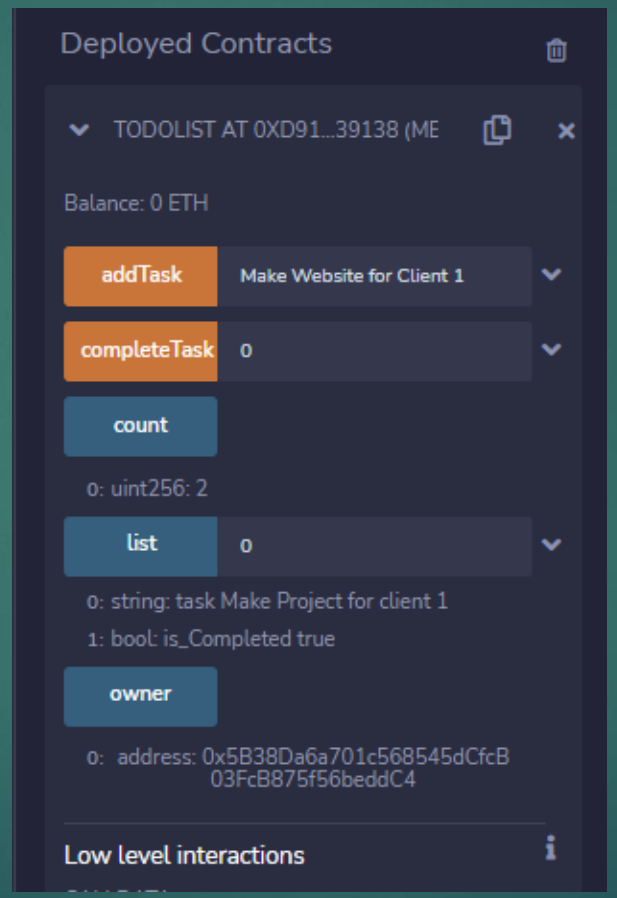

Fig. 9 Complete the task and information

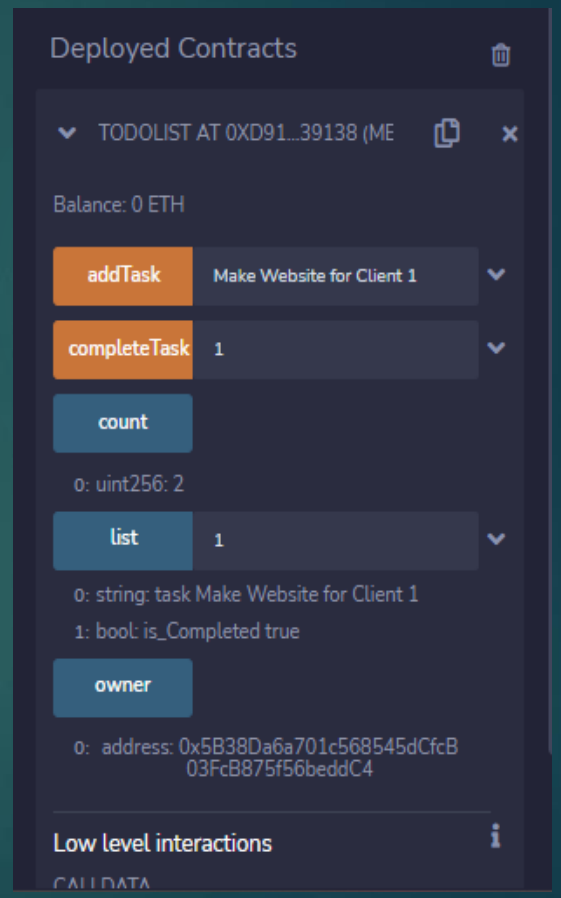

Fig. 10 Complete the 2nd task information

Project 1 :- NFT Marketplace

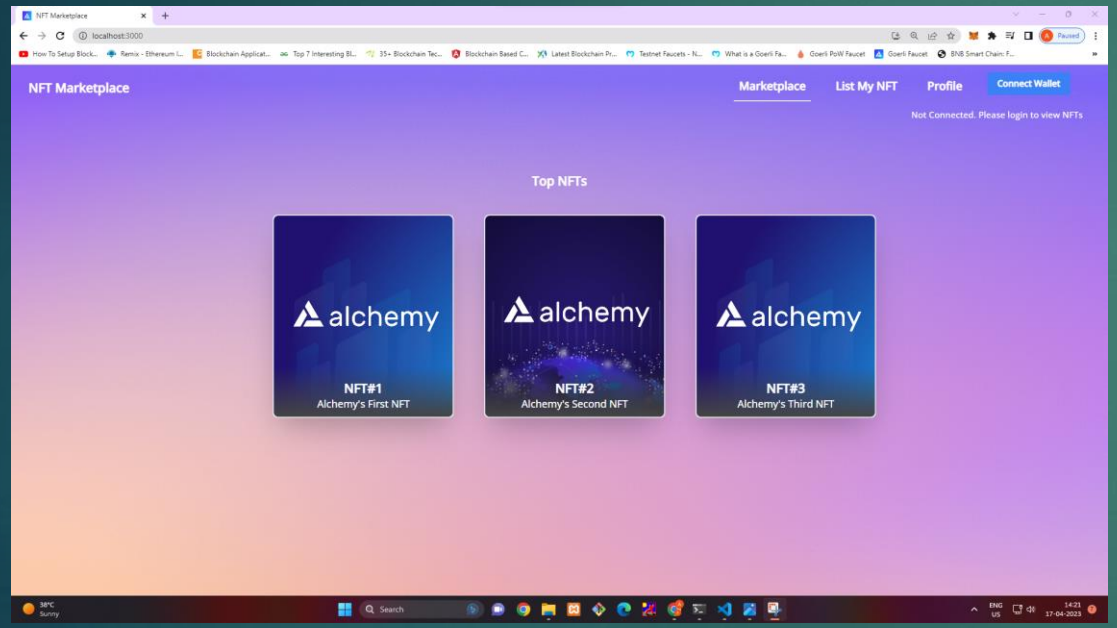

Fig.11 Marketplace Home Page

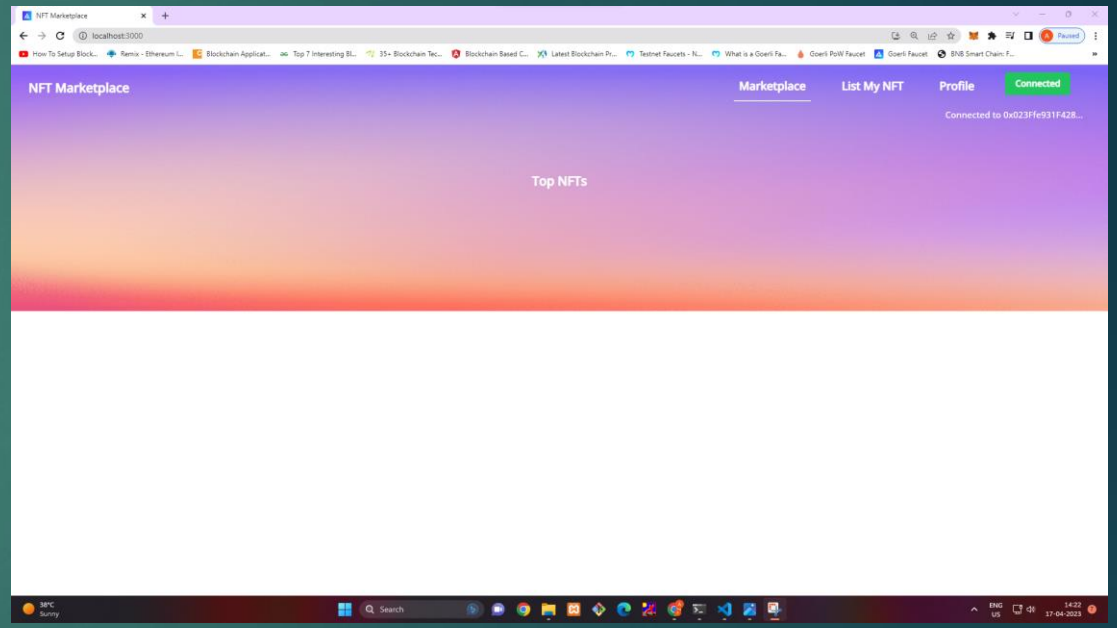

Fig.12 Marketplace Home Page with wallet connected

Project 1 :- NFT Marketplace

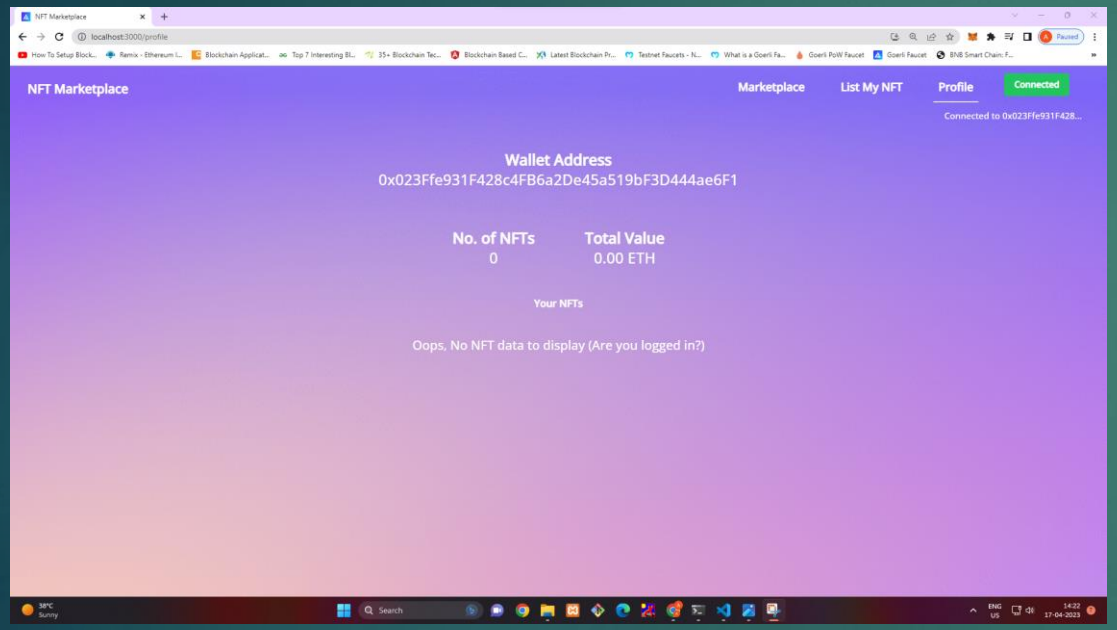

Fig. 13 Profile section in NFT Marketplace

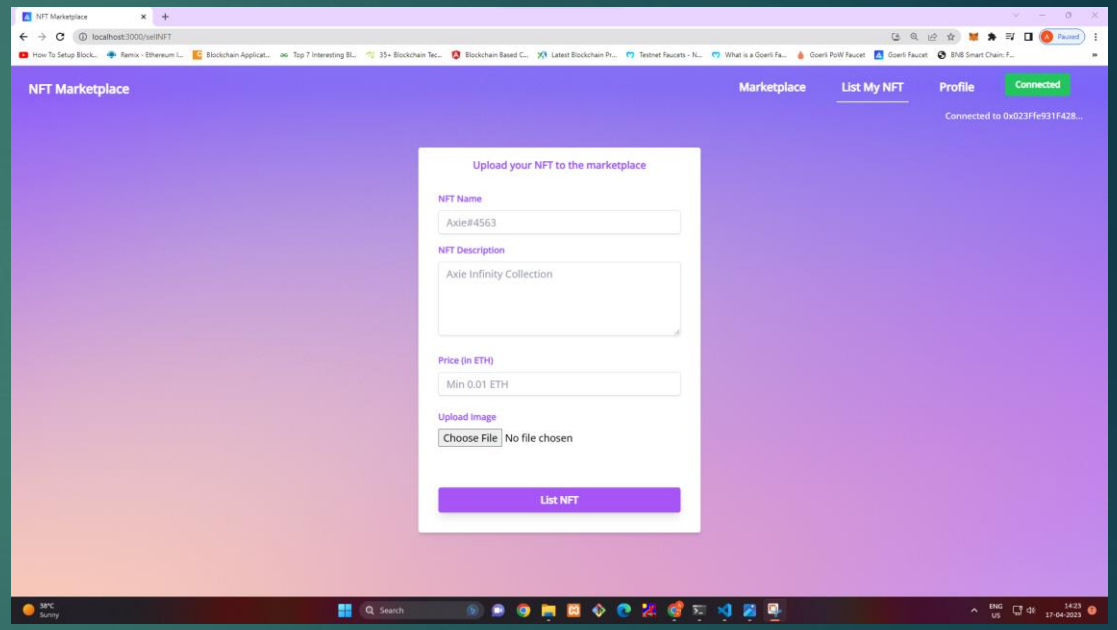

Fig. 14 Listing NFT section in Marketplace

Project 1 :- NFT Marketplace

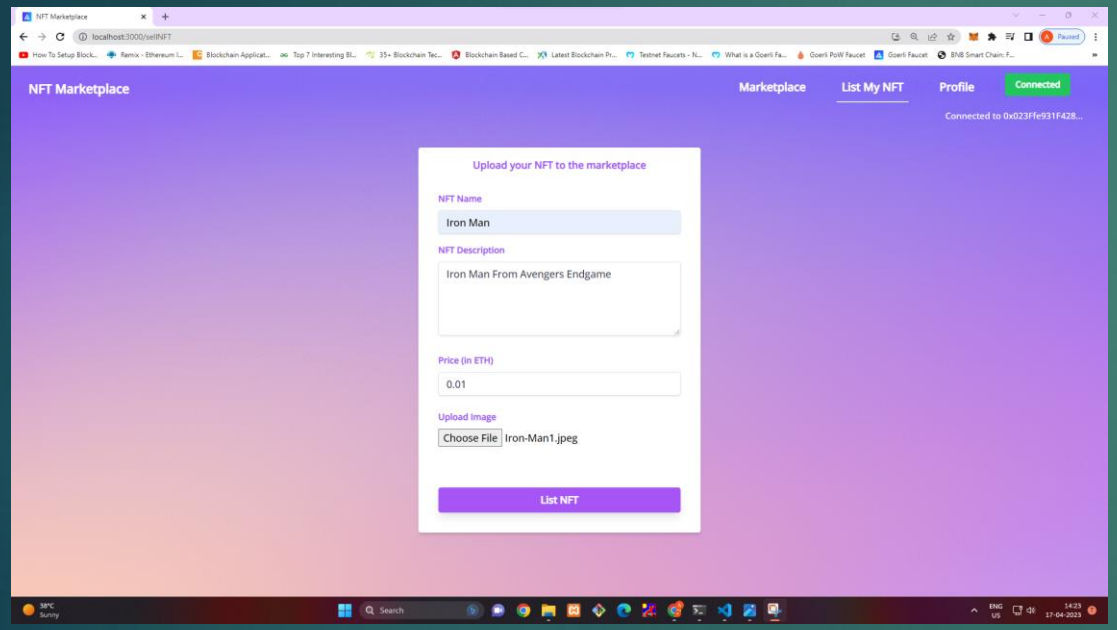

Fig. 15 Listing section after fill the details

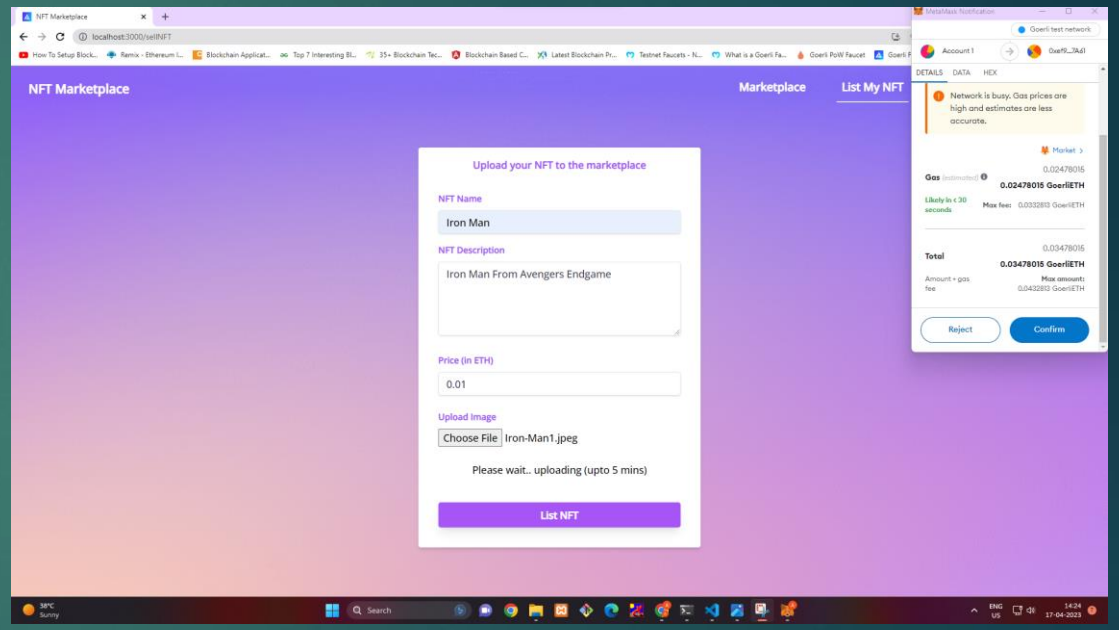

Fig. 16 Transaction For Listing NFT

Project 1 :- NFT Marketplace

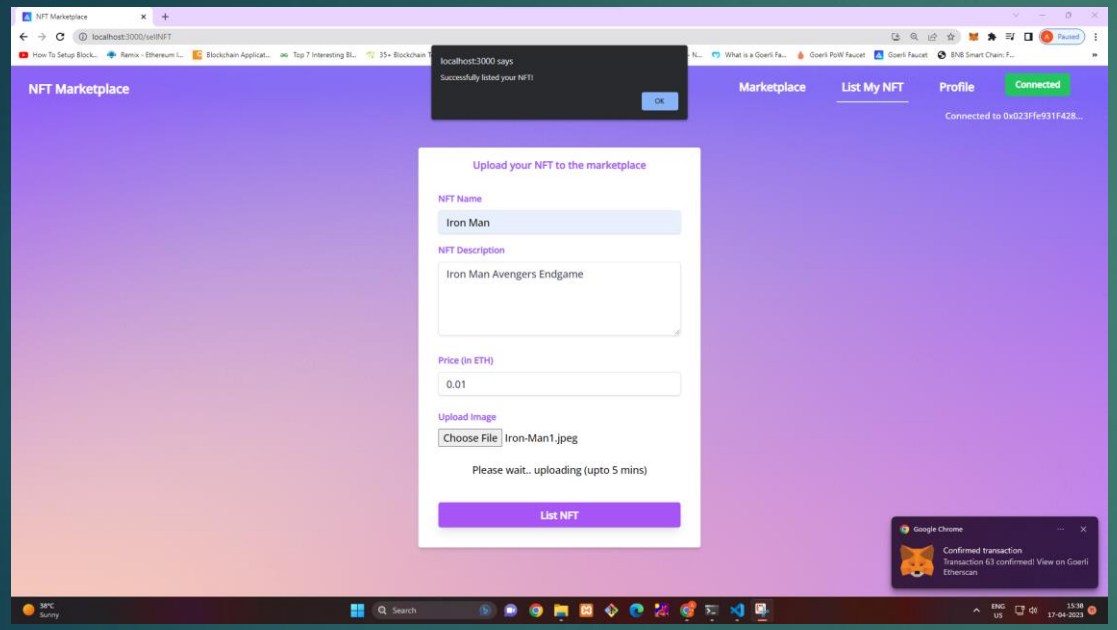

Fig. 17 Listed NFT Successfully Message display

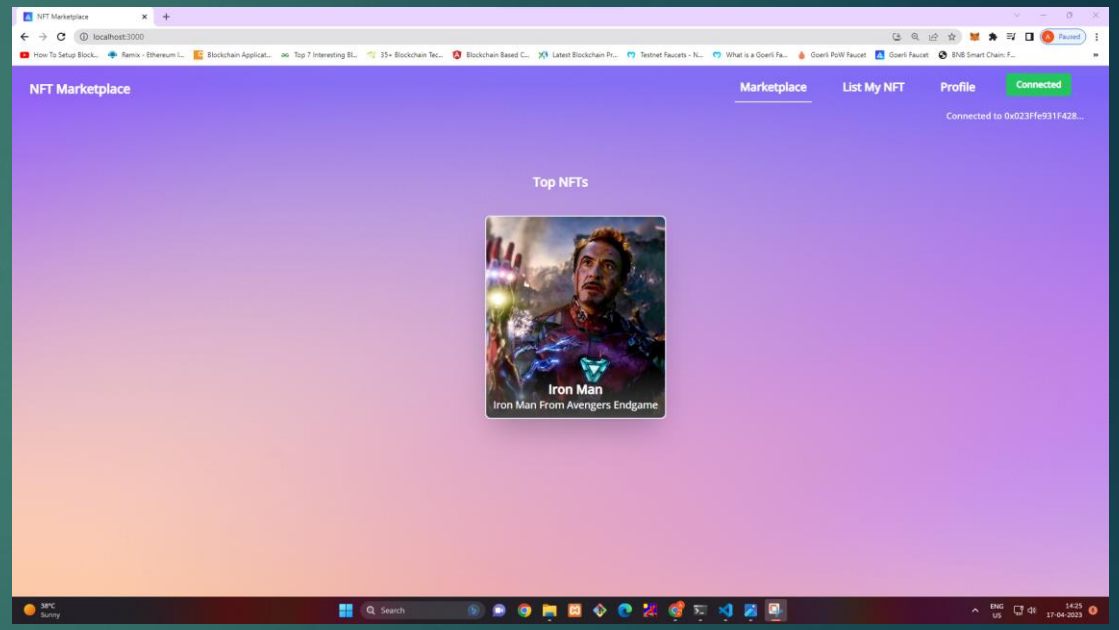

Fig. 18 NFT listed on Marketplace

Project 1 :- NFT Marketplace

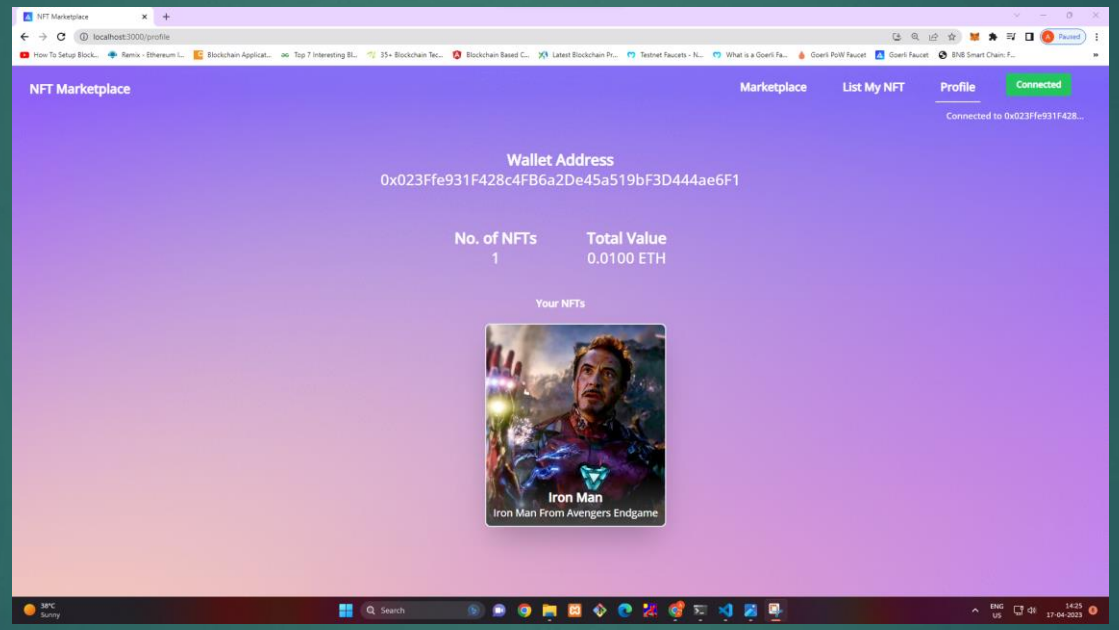

Fig. 19 Show NFT details in profile

Project 2 :- Dappazon

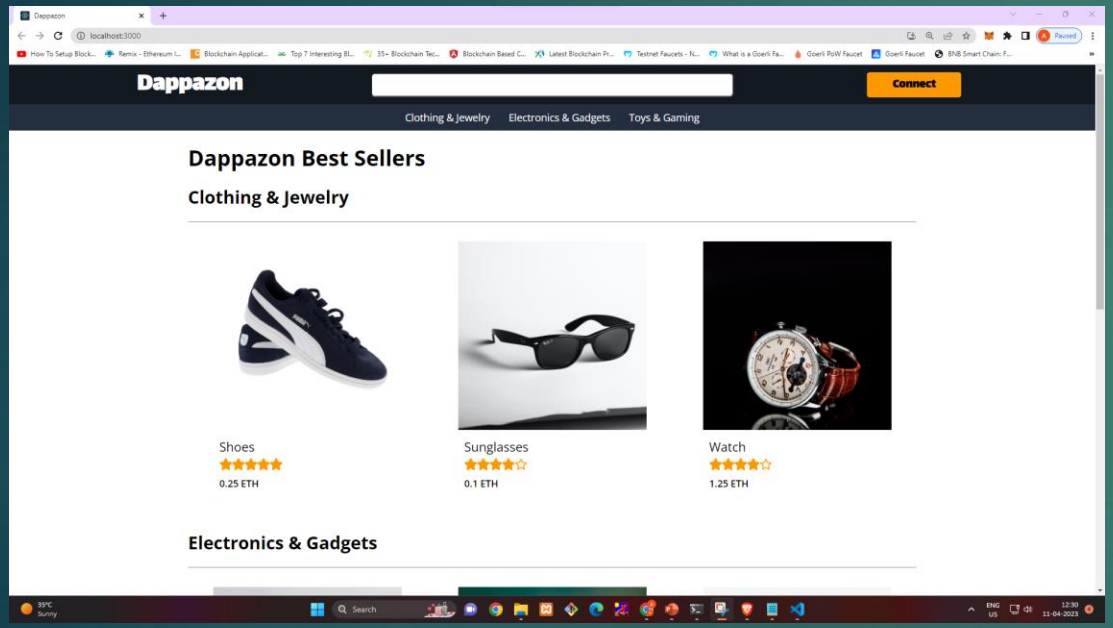

Fig.20 Dappazon

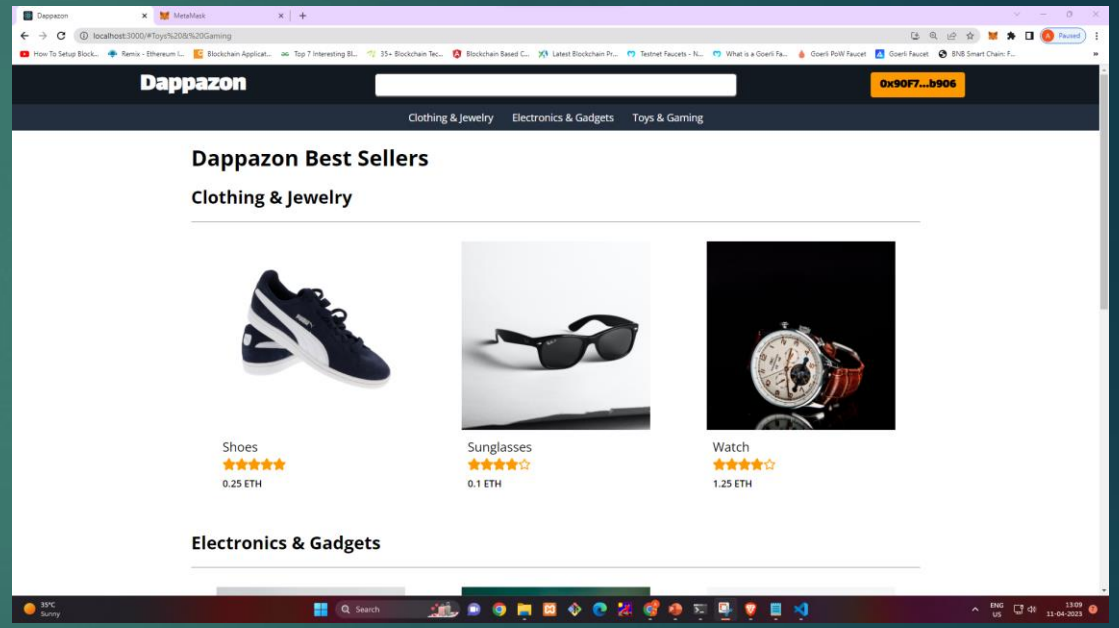

Fig.21 Dappazon after user connect with wallet

Project 2 :- Dappazon

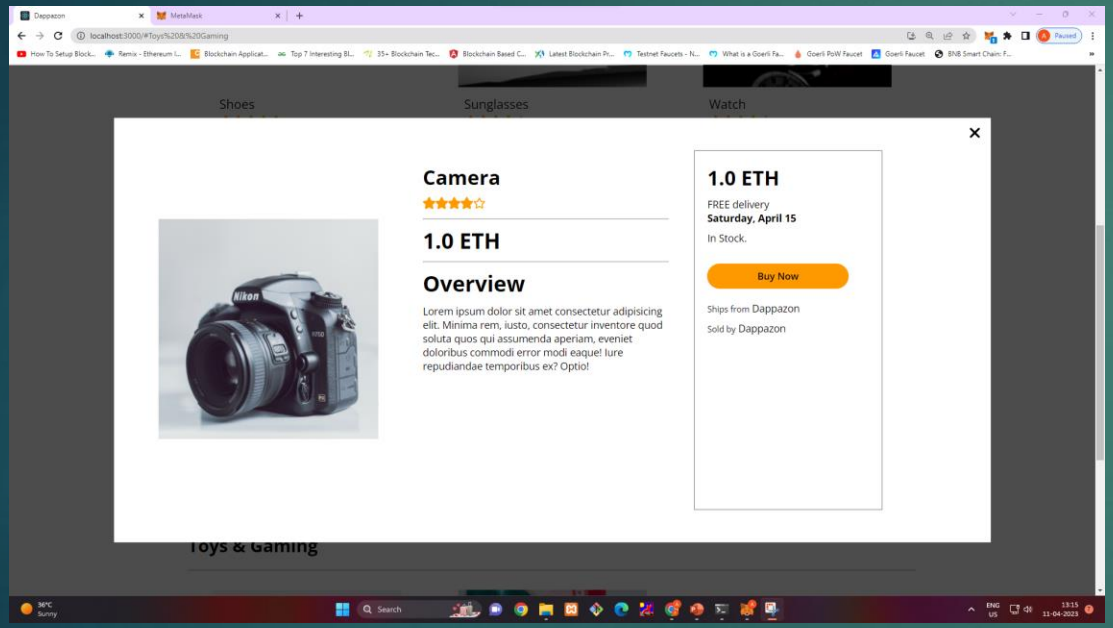

Fig.22 Products are available for sell

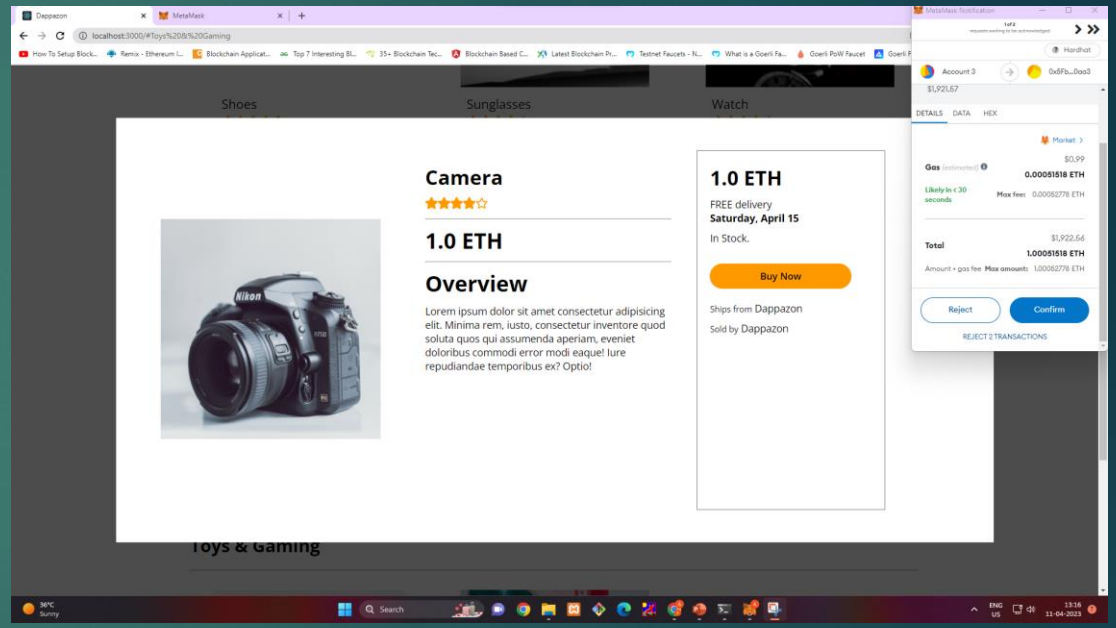

Fig.23 Product has been bought and MetaMask popup appear for transaction

Project 2 :- Dappazon

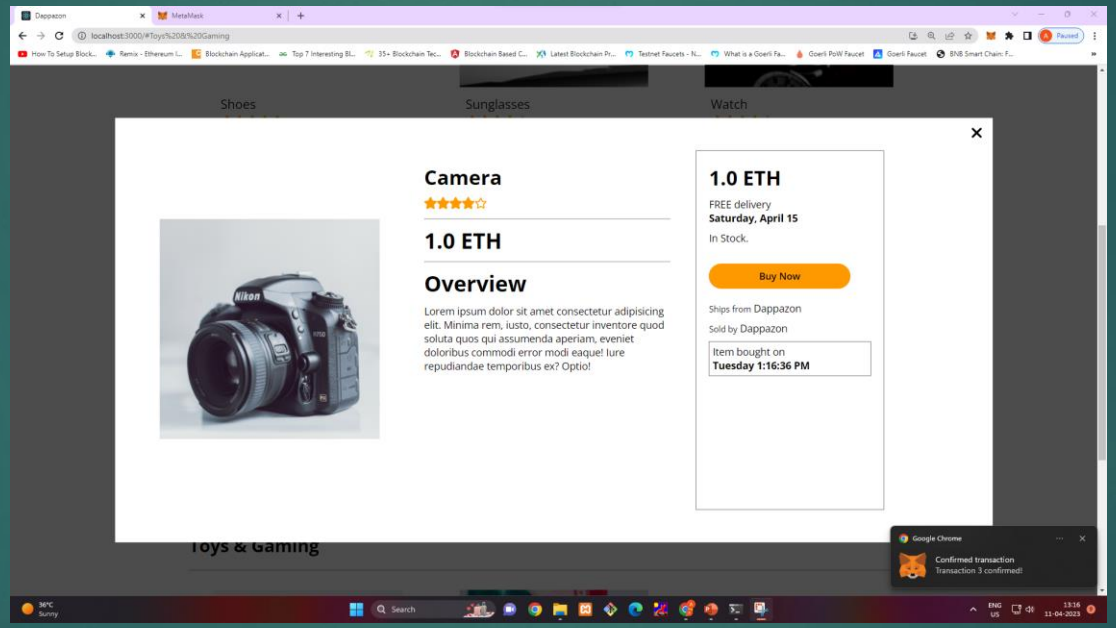

Fig.24 Product was purchased

Project 3 :- Property Sell and Buy

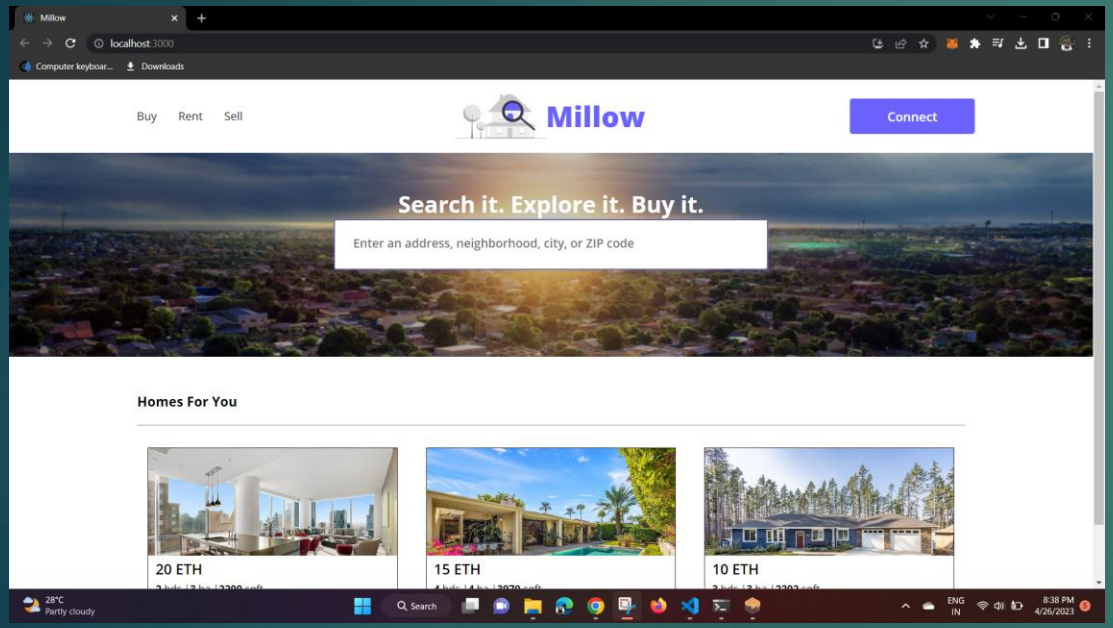

Fig.25 Property Sell and Buy Web3 application

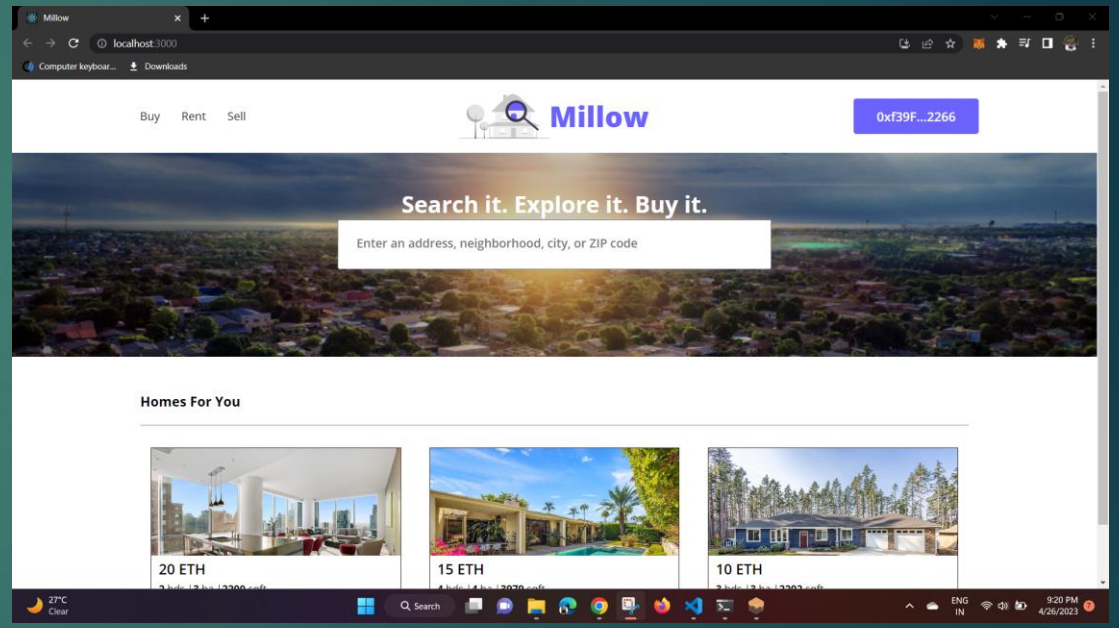

Fig.26 After user connect wallet

Project 3 :- Property Sell and Buy

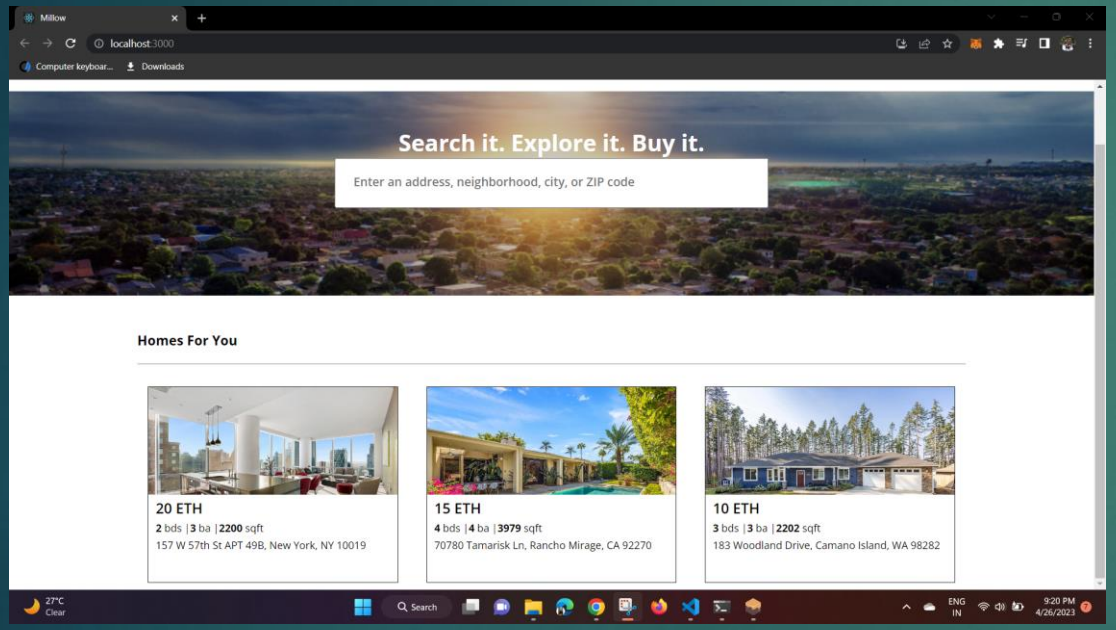

Fig.27 Property are available Sell and Buy

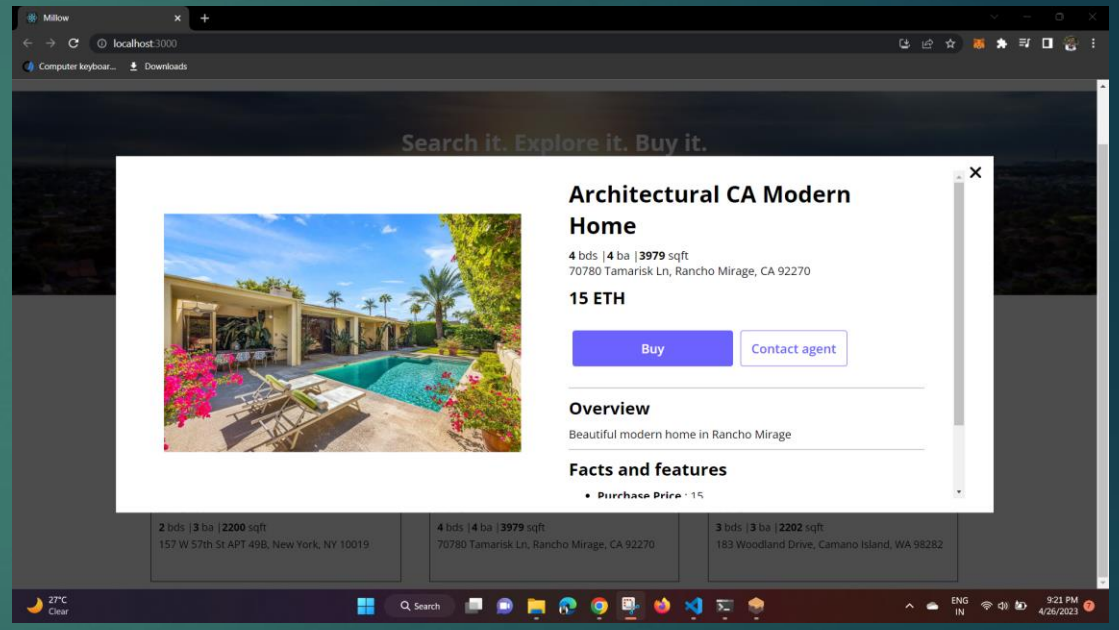

Fig.28 Property is available for buy

Project 3 :- Property Sell and Buy

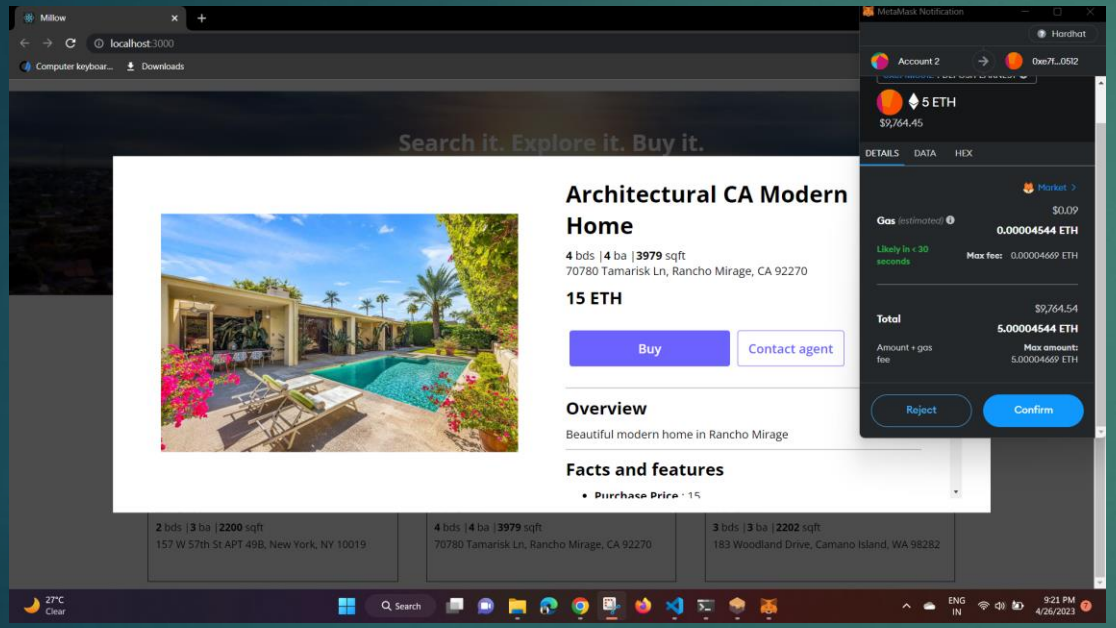

Fig.29 Property is purchase by the user

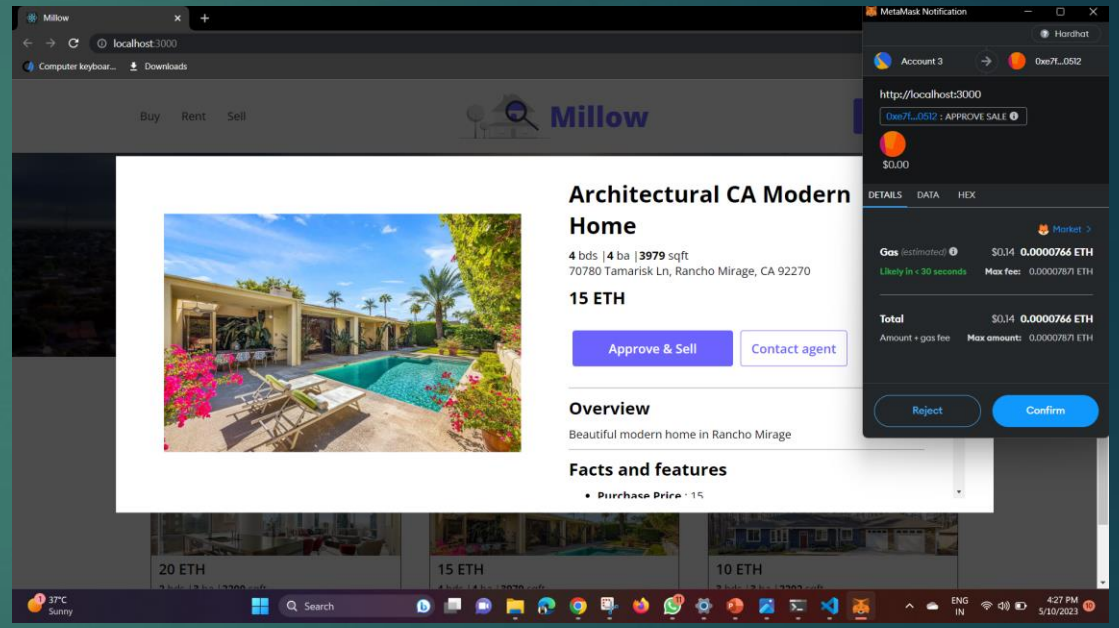

Fig.30 Property is purchased and ready for sell

Project 3 :- Property Sell and Buy

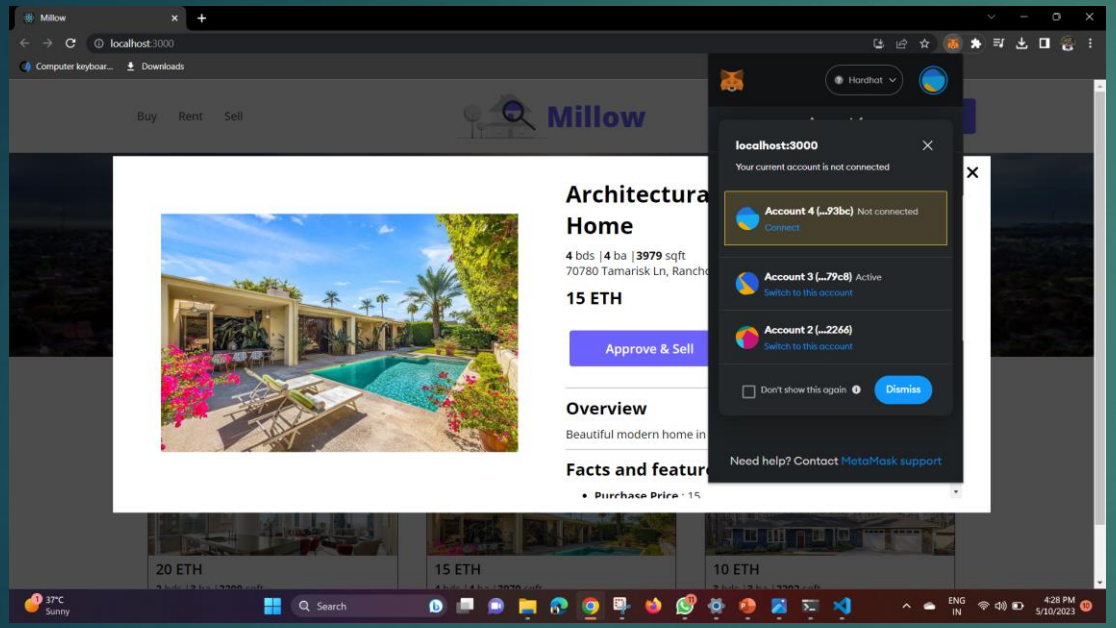

Fig.31 Now changing wallet for inspection

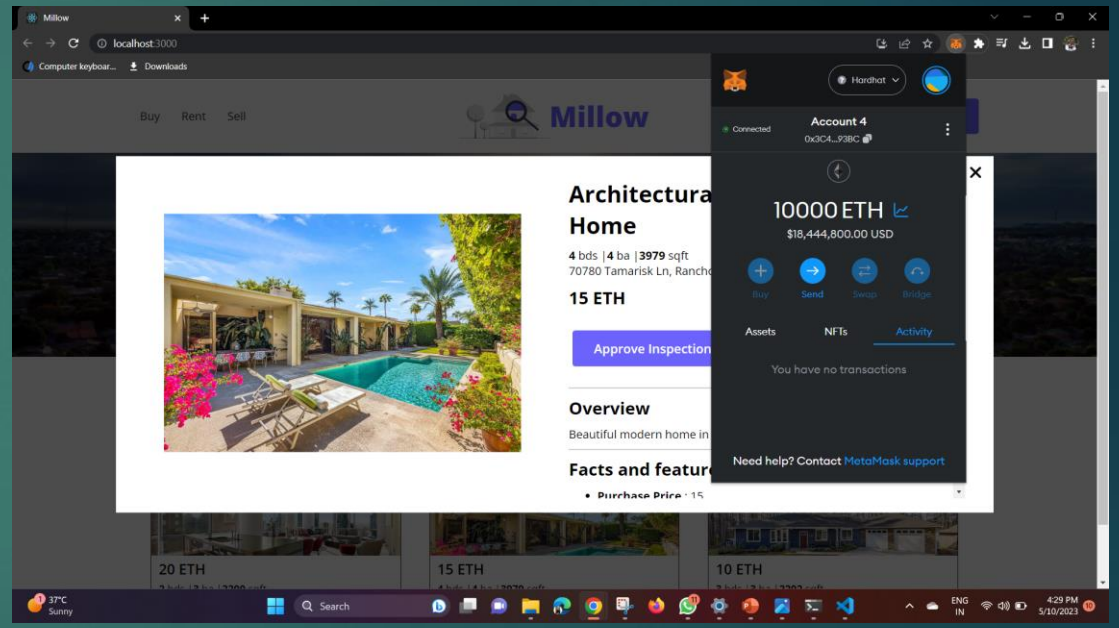

Fig.32 New Account user can do inspection

Project 3 :- Property Sell and Buy

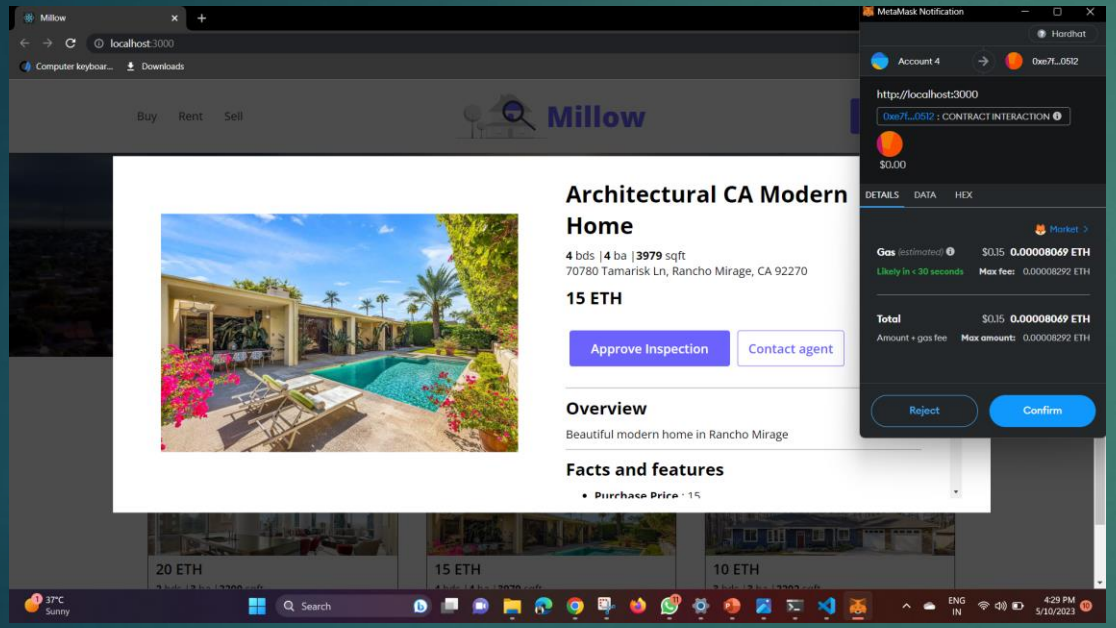

Fig.33 To Confirmation For inspection

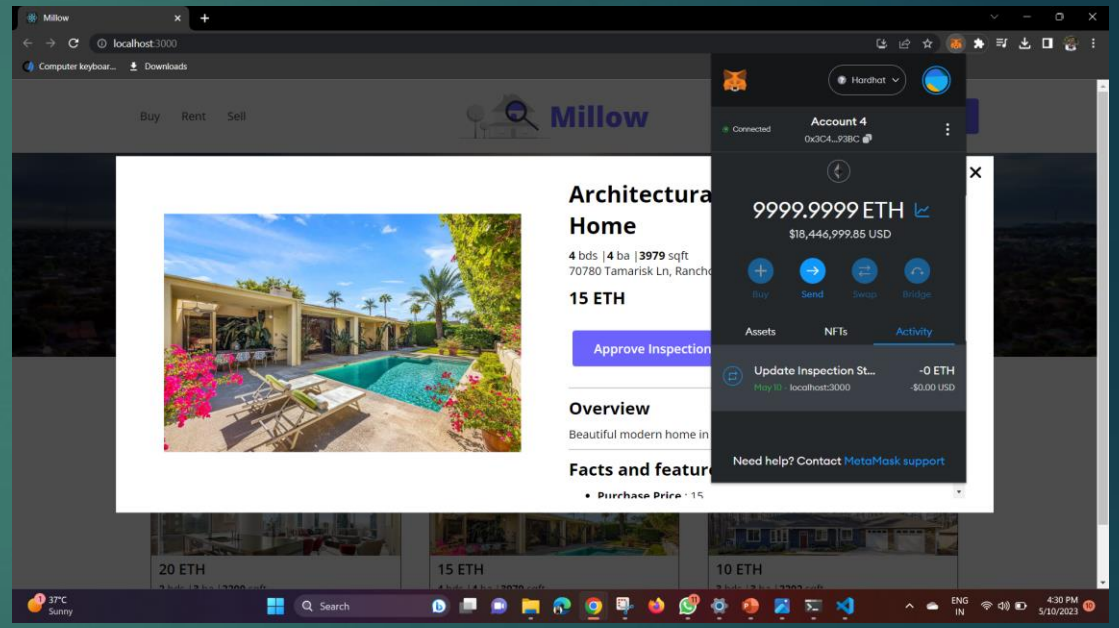

Fig.34 MetaMask show the confirmation ion transaction

Project 3 :- Property Sell and Buy

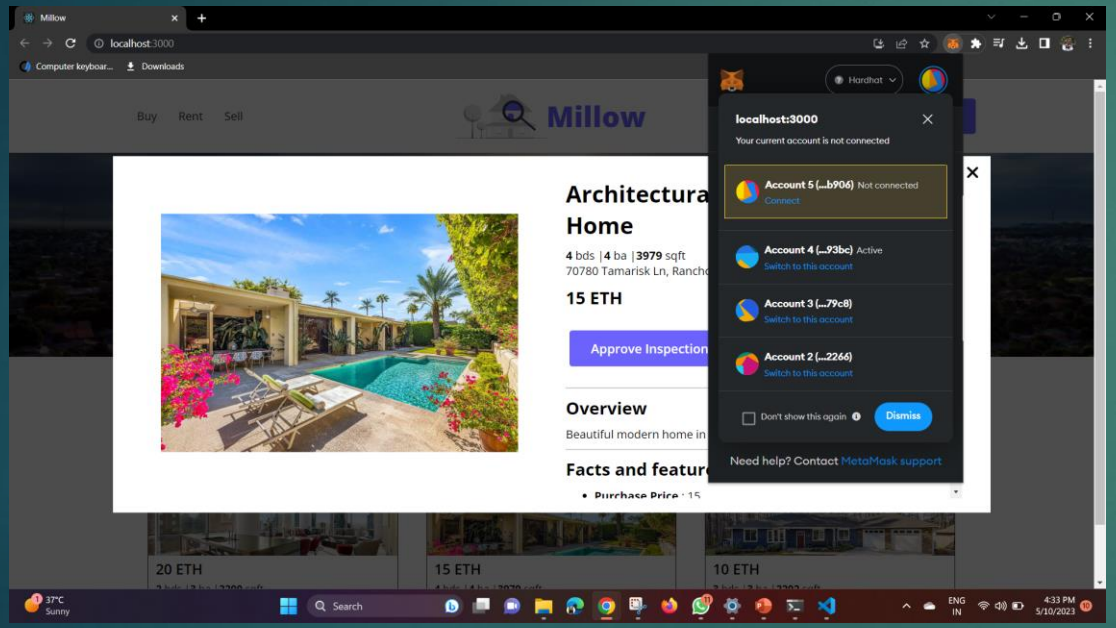

Fig.35 Change the wallet for Lending the property

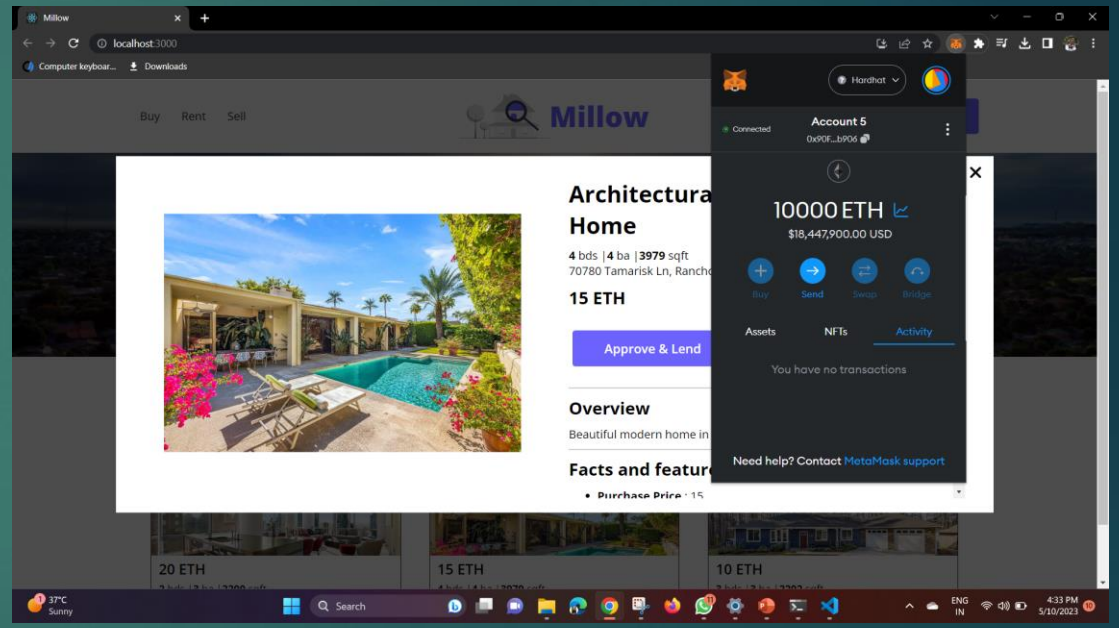

Fig.36 After Change Wallet it show to Lend property

Project 3 :- Property Sell and Buy

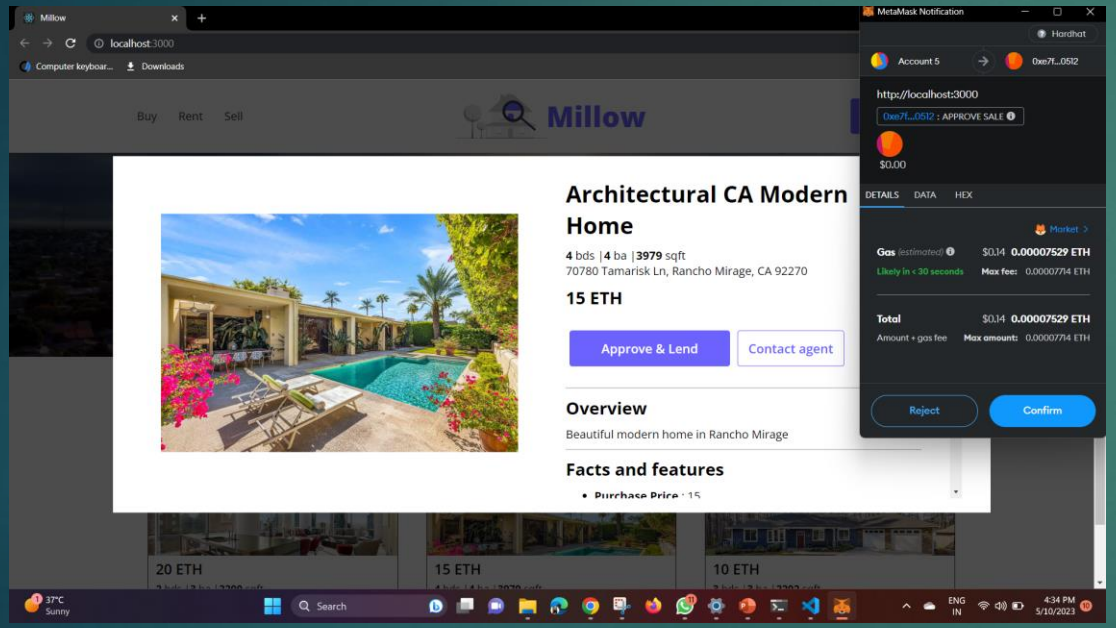

Fig.37 Lend the property

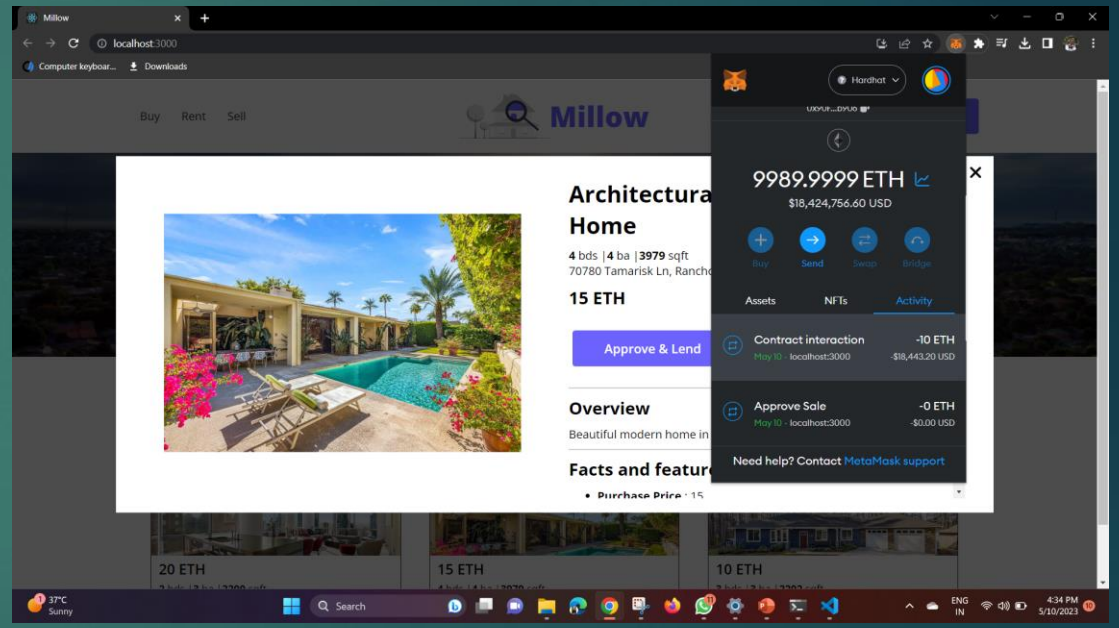

Fig.38 Lend the property transaction is successfully

Conclusion

- In conclusion, my internship experience provided me with a wealth of knowledge and skills that will undoubtedly benefit me in my future endeavors. I had the opportunity to learn about Blockchain Technology and gain hands-on experience working with a talented team at Advait Solution. Through this experience, I was able to hone my soft skills, such as communication, collaboration, and problem-solving, which are integral to success in any professional field. Overall, my internship at Advait Solution was an enriching and rewarding experience that has helped to shape me into a more well-rounded and capable professional.

Future Scope

- Digital Advertising
- Cyber Security
- Blockchain beyond the world of computing
- Supply chain Management
- Blockchain in cloud storage
- Managing World trade
- Governments will provide their digital currencies

References

1) https://web3.career/learn-web3
2) https://en.wikipedia.org/wiki/Blockchain
3) https://soliditylang.org/
4) https://docs.soliditylang.org/en/v0.8.19/
5) https://docs.soliditylang.org/en/v0.8.20-docs/
6) https://trufflesuite.com/docs/
7) https://trufflesuite.com/docs/ganache/
8) https://hardhat.org/tutorial
9) https://www.tutorialspoint.com/blockchain/index.htm

Thank You